SIMATS ENGINEERING

SAVEETHA SCHOOL OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI)

ASSIGNMENT 3 – CO3

DSA03 – NATURAL LANGUAGE PROCESSING

Design, Parse and Probabilistically Rank English Queries for a University Helpdesk Using Context-Free Grammars, Feature Structures and Earley Parsing

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Nature of Assignment	Individual
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Total Marks	50
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MARK SHEET (50 MARKS)

Task	Title	Maximum Marks	Marks Awarded
1	Design a Context-Free Grammar for English Syntax	5	
2	CFG Rules, Derivations and Parse Trees	3	
3	Sentence-Level Constructions	4	
4	Agreement	3	
5	Subcategorization	4	
6	Top-Down Parsing	4	
7	Earley Parsing	5	
8	Feature Structures and Unification	4	
9	Probabilistic Context-Free Grammars	5	
10	Implement and Test the Parsers	3	
11	Compare, Recommend and Defend	2	
12	Day-4 Design Challenge	8	
	TOTAL	50	

MANDATORY SUBMISSION EVIDENCE CHECK

Task	Evidence expected	Status
1	CFG definition, ≥40 rules, ≥60-word lexicon, constituency tests	☐
2	Five left/right derivations, trees, ambiguity examples	☐
3	Declarative, imperative, yes/no and wh-question rules/trees	☐
4	Agreement rule counts and 6 grammatical + 6 ungrammatical tests	☐
5	Five subcategorization frames and accept/reject tests	☐
6	Top-down pseudocode, traces, recursion and complexity	☐
7	Earley chart, operations, back-pointers and complexity	☐
8	Feature matrices, three unifications, feature grammar	☐
9	PCFG estimates, tree probabilities and CKY/Viterbi	☐
10	Python code, 30-sentence evaluation, runtime/state table and plot	☐
11	Parser comparison and deployment recommendation	☐
12A–D	Impact, original design evaluation, adaptation and defence	☐

Evidence note: the assignment asks for a 30-sentence test set, a mini-treebank of at least 20 parsed sentences, and measured parser states/runtime. No annotated treebank or execution log was supplied with the question. The examples and probabilities in this document are explicitly illustrative; replace all fields marked “Measure/record” with results from running the supplied grammar and code before final submission.

TASK 1 – DESIGN A CONTEXT-FREE GRAMMAR FOR ENGLISH SYNTAX (5 MARKS)

1.1 Formal definition

A context-free grammar is G=(N, Σ, R, S), where N is a finite set of non-terminal symbols, Σ is a finite terminal vocabulary, R is a set of production rules A→α with A∈N and α∈(N∪Σ)*, and S is the start symbol. The helpdesk grammar accepts a controlled fragment of English questions and commands about examinations, fees, hostels, library, timetable and placements.

Component	Definition in this grammar
N	S, DECL, IMP, YNQ, WHQ, NP, VP, PP, AP, AdvP, Det, N, Npl, Nsg, Pron, V, Vt, Vi, Vd, Vc, Vp, Aux, Modal, Prep, Conj, Rel, CP, QP, Num, ProperN, etc.
Σ	The words in the lexicon in section 1.3, plus punctuation where retained.
R	The phrase-structure rules R1–R48 below.
S	Start symbol representing a complete helpdesk utterance.

1.2 Phrase-structure rules (48 rules)

Rule	Production
R1	S → DECL
R2	S → IMP
R3	S → YNQ
R4	S → WHQ
R5	DECL → NP VP
R6	IMP → VP
R7	IMP → Please VP
R8	YNQ → Aux NP VP
R9	YNQ → Modal NP VP
R10	WHQ → WhSubj VP
R11	WHQ → WhObj Aux NP VP
R12	WHQ → WhAdv Aux NP VP
R13	NP → Pron
R14	NP → ProperN
R15	NP → Det Nsg
R16	NP → Det Npl
R17	NP → Det AP Nsg
R18	NP → Det AP Npl
R19	NP → Num Npl
R20	NP → NP PP
R21	NP → NP RelClause
R22	NP → NP Conj NP
R23	NP → Poss NP
R24	VP → Vi
R25	VP → Vt NP
R26	VP → Vd NP NP
R27	VP → Vc CP
R28	VP → Vp PP
R29	VP → Aux VP
R30	VP → Modal VP
R31	VP → VP PP
R32	VP → VP AdvP
R33	VP → VP Conj VP
R34	VP → Vt NP PP
R35	PP → Prep NP
R36	AP → Adj
R37	AP → AdvP Adj
R38	AdvP → Adv
R39	AdvP → AdvP Adv
R40	CP → that DECL
R41	CP → whether DECL
R42	RelClause → that VP
R43	RelClause → who VP
R44	RelClause → which VP
R45	WhObj → what
R46	WhObj → which NP
R47	WhAdv → when
R48	WhAdv → where

Recursive rules such as NP→NP PP and VP→VP PP permit arbitrarily long sequences of modifiers in principle. For example, “the form near the office beside the library…” can be extended repeatedly. This is a finite grammar generating an unbounded language.

1.3 Lexicon (over 60 words)

Lexical category	Example entries
Det	the, a, an, this, that, these, those, my, your, each
Pron	I, we, you, he, she, they, me, us, him, her, them, it
Nsg	student, teacher, exam, timetable, hostel, fee, form, room, library, notice, receipt, portal, placement, seminar, result, course, book, desk, department, scholarship
Npl	students, teachers, exams, timetables, hostels, fees, forms, rooms, libraries, notices, receipts, portals, placements, seminars, results, courses, books, desks, departments, scholarships
ProperN	Chennai, Vellore, Saveetha, SIMATS, Monday, Friday
Vi	arrive, depart, crash, sleep, wait, graduate
Vt	submit, check, book, cancel, print, download, open, pay, read, update, attend
Vd	send, give, show, email, tell
Vc	know, say, confirm, report, explain
Vp	apply, depend, wait, register
Aux/Modal	do, does, did, is, are, was, were, have, has, can, could, will, would, should
Prep	to, from, near, in, on, at, for, with, by, before, after, beside
Adj/Adv	late, available, postponed, new, current, online, urgent, final, quickly, tomorrow, today
Conj/Wh	and, or, but, that, whether, who, which, what, when, where, how, please

The lexicon contains more than 100 surface forms. The grammar and lexicon are separated: rules encode structural combinations, while the lexicon records which words can occupy each category. Fine-grained categories such as Vt and Vi are used because verbs differ in complement requirements.

1.4 Constituency, head and phrasal category

Test	Example	Result
Substitution	“The student submitted the form.” Replace “the student” with “they”.	The phrase is an NP because it can be replaced by a pronoun.
Movement	“Near the library, the student booked a room.” Move “near the library” to the sentence front.	The PP behaves as a constituent.
Coordination	“the timetable and the examination notice”	Two NPs coordinate as one larger NP.
Head	“the very late examination”	Noun examination is the head of NP; adjective phrase modifies it.
Head	“book a room near the library”	Verb book is the head of VP; NP is its object.

A constituent is a group of words functioning as a unit. The head determines the phrase’s central category and many of its grammatical properties: N heads NP, V heads VP, and P heads PP. The grammar is context-free because each rule rewrites one non-terminal independently of its surrounding context. Long-distance agreement and cross-serial dependencies can strain a plain CFG; feature unification or mildly context-sensitive formalisms are often used for such phenomena.

TASK 2 – CFG RULES, DERIVATIONS AND PARSE TREES (3 MARKS)

The following five short helpdesk sentences are used for worked derivations. Bracketed notation gives the tree; the ASCII tree gives its hierarchy. Heights count edges from root to deepest terminal; node counts include non-terminals and terminals.

2.1 Worked sentence A

The student submits the form.

Bracketed tree: S(DECL(NP(Det the,Nsg student),VP(Vt submits,NP(Det the,Nsg form))))

```
S
├─ DECL
│  ├─ NP: the student
│  └─ VP
│     ├─ V: submits
│     └─ NP: the form
```

Height 5; nodes 11; rule applications 7

2.2 Worked sentence B

Students book rooms.

Bracketed tree: S(DECL(NP(Npl Students),VP(Vt book,NP(Npl rooms))))

```
S
└─ DECL
   ├─ NP: Students
   └─ VP: book rooms
```

Height 4; nodes 7; rule applications 5

2.3 Worked sentence C

Please submit the form.

Bracketed tree: S(IMP(Please,VP(Vt submit,NP(Det the,Nsg form))))

```
S
└─ IMP
   ├─ Please
   └─ VP: submit the form
```

Height 5; nodes 7; rule applications 5

2.4 Worked sentence D

Does the student book a room?

Bracketed tree: S(YNQ(Aux Does,NP(Det the,Nsg student),VP(Vt book,NP(Det a,Nsg room))))

```
S
└─ YNQ
   ├─ Aux: Does
   ├─ NP: the student
   └─ VP: book a room
```

Height 5; nodes 11; rule applications 7

2.5 Worked sentence E

Where is the library?

Bracketed tree: S(WHQ(WhAdv where,Aux is,NP(Det the,Nsg library)))

```
S
└─ WHQ
   ├─ WhAdv: Where
   ├─ Aux: is
   └─ NP: the library
```

Height 4; nodes 8; rule applications 5

2.2 Leftmost and rightmost derivation example

Sentence: “The student submits the form.” For compactness, use S→DECL, DECL→NP VP, NP→Det Nsg, VP→Vt NP.

Leftmost: S ⇒ DECL ⇒ NP VP ⇒ Det Nsg VP ⇒ the Nsg VP ⇒ the student VP ⇒ the student Vt NP ⇒ the student submits NP ⇒ the student submits Det Nsg ⇒ the student submits the Nsg ⇒ the student submits the form.

Rightmost: S ⇒ DECL ⇒ NP VP ⇒ NP Vt NP ⇒ NP Vt Det Nsg ⇒ NP Vt Det form ⇒ NP Vt the form ⇒ Det Nsg Vt the form ⇒ Det student submits the form ⇒ the student submits the form. At each step, the rightmost non-terminal is expanded.

Both derivations produce the same parse tree. A derivation is an ordered sequence of rule applications; a tree records the hierarchical structure. Different expansion orders can generate the same tree.

2.3 Structural ambiguity

Sentence 1: “Show the students with the late fee notice.”

Tree A – PP attaches to NP: [S [VP Show [NP [NP the students] [PP with [NP the late fee notice]]]]]. Meaning: show the students who have/are associated with the notice.

Tree B – PP attaches to VP: [S [VP [VP Show [NP the students]] [PP with [NP the late fee notice]]]]. Meaning: use the notice as the instrument or accompanying material when showing the students.

Sentence 2: “Book the room near the library for the seminar.” The PP “near the library” may attach to NP room (which room?) or VP book (where to book?). The PP “for the seminar” may also attach to VP or NP depending on intended reading. The number of valid trees increases as attachment sites multiply.

2.4 Membership test

A recogniser tests whether a string belongs to L(G) by attempting to derive it from S, or by chart parsing (Earley) / dynamic programming (CYK after CNF conversion). If no complete parse spans the full input, the string is not generated by the grammar. A recogniser does not automatically determine whether the sentence is factually true or pragmatically appropriate; coverage is limited by the grammar and lexicon.

TASK 3 – SENTENCE-LEVEL CONSTRUCTIONS (4 MARKS)

Construction	Representative rules	Example
Declarative	S→DECL; DECL→NP VP	The student submits the form.
Imperative	S→IMP; IMP→VP | Please VP	Please submit the form.
Yes/no question	S→YNQ; YNQ→Aux NP VP | Modal NP VP	Does the student book a room?
Wh-subject question	S→WHQ; WHQ→WhSubj VP	Who booked the room?
Wh-object question	WHQ→WhObj Aux NP VP	Which room did the student book?
Wh-adverb question	WHQ→WhAdv Aux NP VP	When will the exams begin?

Auxiliary inversion moves the finite auxiliary before the subject in yes/no questions: “The student is ready” → “Is the student ready?” Wh-movement places a wh-phrase at the beginning: “The student booked which room?” → “Which room did the student book?” A gap or trace can be represented in a more detailed grammar: WHQ→WhObj Aux NP VP[gap].

3.1 Subordinate and relative clauses

Rule	Example
CP→that DECL	I know that the fee is paid.
CP→whether DECL	Tell me whether the hostel is open.
NP→NP RelClause	the exams that were postponed
RelClause→that VP	the form that arrived yesterday
RelClause→who VP	the teacher who approved it

3.2 Parse sketches

Type	Bracketed representation	Structural note
Declarative	[S [DECL [NP the student] [VP submits [NP the form]]]]	Canonical NP–VP order.
Imperative	[S [IMP [VP submit [NP the form]]]]	Subject is understood as you.
Yes/no	[S [YNQ Does [NP the student] [VP book [NP a room]]]]	Finite auxiliary precedes subject.
Wh-object	[S [WHQ [WhNP which room] did [NP the student] [VP book __]]]	Wh phrase is linked to an object gap.

Coordination is represented by NP→NP Conj NP and VP→VP Conj VP. Long-distance dependencies become cumbersome when a wh-phrase must be linked to a gap inside several embedded clauses. Plain CFGs can encode selected patterns, but feature threading or a more expressive grammar is more economical.

Every construction adds rules and creates potential ambiguity. Questions add auxiliary inversion and wh-position alternatives; subordinate and relative clauses add recursive embedding; coordination permits multiple grouping structures. A controlled fragment should limit optional recursion and use features to constrain combinations.

TASK 4 – AGREEMENT (3 MARKS)

The grammar must enforce (1) subject–verb agreement in person and number, (2) determiner–noun number agreement, and (3) pronoun case. Without constraints, it may accept "*The students submits the form" or "*Me submits the form."

4.1 Category multiplication approach

Rule family	Example split
Sentence	S→NP-SG VP-SG | NP-PL VP-PL
NP	NP-SG→Det-SG N-SG; NP-PL→Det-PL N-PL
Pronoun	NP-SUBJ→I|we|he|she|they; NP-OBJ→me|us|him|her|them
Verb	VP-SG→V-SG NP; VP-PL→V-PL NP

For a minimal number-only grammar, splitting S, NP and VP creates at least 2 sentence rules, 2 NP rules and 2 VP rules = 6 agreement-specific rules, before lexical and other phrase rules are counted. If person (3), number (2), case (2) and gender (3) are combined, the feature bundle has 3×2×2×3=36 possible combinations. A naïve fully cross-product grammar can require up to 36 paired subject/verb patterns at the sentence level, and category variants multiply throughout the grammar.

A more economical feature grammar stores values such as [NUM=sg, PER=3] on NP and VP, and requires equality at S→NP VP. The grammar rules remain general; unification checks compatibility.

4.2 Test set

Input	Gold grammaticality	Baseline CFG	Agreement-aware
The student submits the form.	Grammatical	Accept	Accept
The students submit the forms.	Grammatical	Accept	Accept
I submit the form.	Grammatical	Accept	Accept
They book a room.	Grammatical	Accept	Accept
Does the student pay the fee?	Grammatical	Accept	Accept
Please print the timetable.	Grammatical	Accept	Accept
The students submits the form.	Ungrammatical	May accept	Reject
The student submit the form.	Ungrammatical	May accept	Reject
These student books a room.	Ungrammatical	May accept	Reject
Me submit the form.	Ungrammatical	May accept	Reject
They is in the library.	Ungrammatical	May accept	Reject
The forms is ready.	Ungrammatical	May accept	Reject

The baseline column is described as "may accept" because a grammar without agreement constraints can license these strings if it uses generic NP and VP categories. The agreement-aware grammar rejects them by feature mismatch.

TASK 5 – SUBCATEGORIZATION (4 MARKS)

Subcategorization (valence) specifies the number and type of complements a lexical head requires or permits. A verb's frame determines whether it can appear alone, with an NP, two NPs, a clause or a PP.

Frame	Helpdesk verb	Pattern	Example
Intransitive (Vi)	crash, arrive, depart	VP→Vi	The portal crashed.
Transitive (Vt)	submit, check, book	VP→Vt NP	Submit the form.
Ditransitive (Vd)	send, give, show	VP→Vd NP NP	Send me the receipt.
Sentential complement (Vc)	know, confirm, explain	VP→Vc CP	I know that the fee is paid.
PP complement (Vp)	apply, depend, wait	VP→Vp PP	Apply for a hostel room.

5.1 Distinct verb categories

The lexicon initially assigns verbs to V-intr, V-trans, V-ditrans, V-comp and V-PP. The grammar then licenses VP→Vi; VP→Vt NP; VP→Vd NP NP; VP→Vc CP; VP→Vp PP. This prevents a transitive verb from appearing without its required object and prevents an intransitive verb from taking an unsupported object.

Drawback: verbs with multiple frames must be listed in several categories, the lexicon becomes repetitive, and category multiplication becomes difficult when optional complements and semantic roles are added. Feature-valued SUBCAT lists offer a more compact alternative.

5.2 Complement versus adjunct

Test	Complement	Adjunct
Selection	Selected by the verb; e.g., apply for a room.	Adds optional circumstance; e.g., yesterday.
Omission	May make a selected use incomplete: "*Submit" (in this intended transitive use).	Usually removable: "The student submitted the form yesterday."
Multiplicity	Usually limited by the verb frame.	Can often repeat: in the library, after class, yesterday.
Example	wait for the result (selected PP frame)	wait patiently (manner adjunct)

5.3 Over/under-complementation

Sentence	Expected result	Reason
The portal crashed.	Accept	Intransitive verb, no object.
Submit the form.	Accept	Transitive verb with NP object.
Send me the receipt.	Accept	Ditransitive frame.
Tell me that the fee is paid.	Accept	Ditransitive/verb + clause complement.
Apply for a hostel room.	Accept	PP complement.
*Submit.	Reject in required-object reading	Transitive complement missing.
*The portal crashed the form.	Reject	Intransitive verb given an object.
*The student sent.	Reject in ditransitive use	Required recipient/theme missing.

Some verbs are polyvalent: "book" can be transitive ("book a room") or take an additional PP ("book a room for the seminar"). Multiple frames increase lexical ambiguity and parsing alternatives. A parser should retain all licensed frames and let context/probability rank them.

TASK 6 – TOP-DOWN PARSING (4 MARKS)

6.1 Recursive-descent algorithm

```
parse(symbol, input_position):
  if symbol is a terminal: match input[position]; return next position or failure
  for each production symbol -> rhs in grammar[symbol]:
      try matching rhs from current position
      if every rhs item succeeds: return resulting position/tree
      otherwise backtrack and try next production
  return failure
```

A top-down parser begins with goal S and expands non-terminals until terminals match the input. It is goal-driven and can avoid exploring structures that cannot begin from S, but naïve backtracking may repeat the same work.

6.2 Trace on an unambiguous sentence

Input: “The student submits the form.” (punctuation ignored)

Step	Stack / goal	Action
1	S	Expand S→DECL.
2	DECL	Expand DECL→NP VP.
3	NP VP	Expand NP→Det Nsg.
4	Det Nsg VP	Match “the”; match “student”.
5	VP	Expand VP→Vt NP.
6	Vt NP	Match “submits”.
7	NP	Expand NP→Det Nsg; match “the form”.
8	Input exhausted	Success.

6.3 Trace on an ambiguous PP attachment

Input: “Book the room near the library.” At VP, the parser can select VP→Vt NP and build NP→NP PP, attaching “near the library” to room. Alternatively it can select VP→VP PP, attaching the PP to the booking event. A backtracking parser explores one choice, then returns to the PP attachment point and explores the other.

6.4 Left recursion and remedies

A rule NP→NP PP is left-recursive. A recursive-descent parser that expands NP→NP PP before consuming any input calls NP again indefinitely. Eliminate immediate left recursion by rewriting NP→BaseNP NP-Modifier* (or NP→BaseNP NP-Ext; NP-Ext→PP NP-Ext | ε). This preserves the intended repeated-PP structures, although the tree shape changes and a tree-reconstruction step may be needed.

For k PPs, there are k+1 possible attachment boundaries in a simplified single-head setting; with multiple possible hosts, combinations can grow combinatorially. Naïve top-down parsing may repeatedly parse identical NP/VP spans for alternative choices. Worst-case CFG parsing by exhaustive backtracking can be exponential; memoisation/left-corner filtering reduces redundant exploration.

TASK 7 – EARLEY PARSING (5 MARKS)

7.1 State and operations

An Earley state is [A→α • β, i], where A→αβ is a grammar rule, the dot marks how much has been recognised, and i is the origin chart position where this constituent began. Chart S[j] contains states that have processed the first j input tokens.

Operation	When	Action
PREDICTOR	Dot before non-terminal B	Add each B→•γ state to current chart.
SCANNER	Dot before terminal matching next input word	Advance dot and add state to next chart.
COMPLETER	A state B→γ• is complete	Advance every state in its origin chart that was waiting for B.

7.2 Hand simulation (six words)

Input: "The student submits the fee form today." Tokens: 1 the, 2 student, 3 submits, 4 the, 5 fee, 6 form, 7 today. For clarity, use the following miniature grammar: S→NP VP; NP→Det N; NP→NP N; VP→Vt NP Adv; Det→the; N→student|fee|form; Vt→submits; Adv→today. This grammar recognises the chosen sentence and keeps the chart compact.

Chart	Representative states added (• marks current position)
S[0]	S→•NP VP [0]; NP→•Det N [0]; NP→•NP N [0]; Det→•the [0]
S[1]	Det→the• [0]; NP→Det•N [0]; N→•student [1]; N→•fee [1]; N→•form [1]
S[2]	N→student• [1]; NP→Det N• [0]; S→NP•VP [0]; VP→•Vt NP Adv [2]; Vt→•submits [2]
S[3]	Vt→submits• [2]; VP→Vt•NP Adv [2]; NP→•Det N [3]; Det→•the [3]
S[4]	Det→the• [3]; NP→Det•N [3]; N→•student [4]; N→•fee [4]; N→•form [4]
S[5]	N→fee• [4]; NP→Det N• [3]; NP→NP•N [3]; N→•student [5]; N→•fee [5]; N→•form [5]
S[6]	N→form• [5]; NP→NP N• [3]; VP→Vt NP•Adv [2]; Adv→•today [6]
S[7]	Adv→today• [6]; VP→Vt NP Adv• [2]; S→NP VP• [0]; S→NP VP• [0] completed: accept

The chart above is a compact hand trace showing the principal states; a full implementation may add duplicate-suppressed states and intermediate predictions not repeated in the summary. State notation [origin] is shown in brackets. Scanner advances over terminals; completer advances parent states when NP/VP completes.

7.3 Back-pointers and ambiguity

Store a back-pointer whenever a state is created: scanner pointers identify the consumed terminal; completer pointers identify the completed child and the waiting parent. For "Show the students with the late fee notice", two completed PP attachment paths are retained in a packed parse forest: PP as a child of NP, or PP as a child of VP. Shared subtrees are stored once; alternatives are packed rather than duplicated.

Earley handles left recursion because a state is inserted into a chart only once (or deduplicated by state identity). The algorithm does not recursively call itself without consuming input; chart membership prevents an infinite loop.

7.4 Complexity

For a general CFG, Earley has $O(n^3)$ worst-case time and $O(n^2)$ chart space for input length n; unambiguous or nearly deterministic grammars are often faster in practice. Compare actual number of chart states against top-down recursive calls for the same input; record measured values in Task 10.

TASK 8 – FEATURE STRUCTURES AND UNIFICATION (4 MARKS)

8.1 Feature structures

A feature structure is an attribute–value matrix (AVM). Feature paths navigate nested attributes, e.g., NP.AGR.NUM. Reentrancy means two paths point to the same value/structure, enforcing shared constraints such as subject agreement with finite verb agreement.

Lexical item	Feature structure
student	[CAT=N, NUM=sg, PER=3]
students	[CAT=N, NUM=pl, PER=3]
submits	[CAT=V, FIN=+, AGR=(NUM=sg, PER=3), SUBCAT=<NP>]
submit	[CAT=V, FIN=base, SUBCAT=<NP>]
me	[CAT=PRON, CASE=obj, NUM=sg, PER=1]
I	[CAT=PRON, CASE=subj, NUM=sg, PER=1]
apply	[CAT=V, SUBCAT=<PP(for)>]

8.2 Unification and subsumption

Unification combines compatible feature structures by merging their information. If one structure says NUM=sg and another says NUM=sg, the result retains NUM=sg. If one says NUM=sg and the other NUM=pl, unification fails. A more specific structure subsumes a more general one when it contains all its constraints plus additional compatible information.

Example	Inputs	Result
1 – success	[NUM=sg, PER=3] ⊔ [NUM=sg]	[NUM=sg, PER=3]
2 – success	[CASE=obj] ⊔ [CASE=obj, PER=1]	[CASE=obj, PER=1]
3 – failure	[NUM=sg] ⊔ [NUM=pl]	FAIL: conflicting NUM values

8.3 Feature-based grammar rules

Rule	Feature constraint
S→NP VP	NP.AGR = VP.AGR
NP→Det N	Det.NUM = N.NUM
NP→Pron	Pron.CASE = subj in subject position
VP→Vt NP	V.SUBCAT = <NP>
VP→Vi	V.SUBCAT = <>
VP→Vp PP	V.SUBCAT = <PP(PREP)>
VP→Vc CP	V.SUBCAT = <CP>
YNQ→Aux NP VP	Aux.AGR = NP.AGR; finite VP agreement checked.

For “The student submits the form”, Det the unifies with singular student, NP has AGR=[NUM=sg, PER=3], and submits requires AGR=[NUM=sg, PER=3] and SUBCAT=<NP>. The subject and finite verb unify successfully; the object NP satisfies the transitive complement.

Task 4 becomes one general S→NP VP rule with agreement unification instead of separate SG/PL rules. Task 5 uses a SUBCAT list on the verb rather than a large family of duplicated verb categories. Feature structures reduce rule duplication but add unification cost and implementation complexity.

In Earley parsing, unification is performed when a predicted constituent completes and is combined with a waiting parent. Incompatible states are discarded. The worst-case CFG bound remains cubic for standard Earley; feature unification can increase constants and, for unrestricted feature formalisms, may change theoretical complexity.

TASK 9 – PROBABILISTIC CONTEXT-FREE GRAMMARS (5 MARKS)

9.1 PCFG definition and normalisation

A PCFG assigns each production A→β a conditional probability P(A→β|A). For every left-hand side A, the probabilities of all its alternatives must sum to 1. A parse-tree probability is the product of the probabilities of all rule applications in the tree.

LHS	Rule	Count in illustrative treebank	MLE probability
S	S→DECL	16	16/20 = 0.80
S	S→YNQ	3	3/20 = 0.15
S	S→IMP	1	1/20 = 0.05
NP	NP→Det N	12	12/20 = 0.60
NP	NP→Pron	5	5/20 = 0.25
NP	NP→ProperN	3	3/20 = 0.15
VP	VP→Vt NP	10	10/20 = 0.50
VP	VP→Vi	4	4/20 = 0.20
VP	VP→Vp PP	3	3/20 = 0.15
VP	VP→Vc CP	3	3/20 = 0.15

Counts are illustrative allocations for a 20-tree mini-treebank. For each LHS, probabilities sum to one: S=.80+.15+.05=1; NP=.60+.25+.15=1; VP=.50+.20+.15+.15=1. In an actual experiment, count each rule occurrence in the annotated treebank and divide by the total number of rules expanding that LHS.

MLE: P(A→β|A)=Count(A→β)/ΣγCount(A→γ). Unseen rules receive zero under MLE; use add-α smoothing or a small reserved mass for unseen alternatives, then renormalise each LHS row. Explain the chosen α.

9.2 Ambiguous tree probability

For “Book the room near the library”, compare two simplified structures. Tree A: PP attaches to NP room. Tree B: PP attaches to VP book. Suppose the illustrative PCFG probabilities are: P(S→IMP)=0.20, P(IMP→VP)=1, P(VP→Vt NP)=0.60, P(NP→Det N)=0.70, P(NP→NP PP)=0.20, P(VP→VP PP)=0.10, P(PP→Prep NP)=1, P(Det→the)=1, P(N→room)=0.5, P(N→library)=0.5, P(Prep→near)=1, P(Vt→book)=1.

Tree A probability = .20×1×.60×.70×.20×1×1×1×.5×.5×1 = 0.0042. Tree B probability = .20×1×.10×.60×.70×1×1×.5×.5×1 = 0.0021. Thus Tree A is preferred in this illustrative parameterisation. Sentence probability is the sum over its distinct trees: 0.0042+0.0021=0.0063.

The values are demonstration probabilities, not estimated from the user's real treebank. Replace them with counts from at least 20 annotated sentences.

9.3 Probabilistic CKY / Viterbi

```
for span_length = 1..n:
  for each span (i,j):
    initialise lexical/preterminal entries
    for each binary rule A -> B C:
      for each split k between i and j:
        score = P(A->BC) * best[i,k,B] * best[k,j,C]
        if score > best[i,j,A]: store score and backpointer
return best[0,n,S]
```

Convert longer CFG rules to Chomsky Normal Form (CNF): rules are A→BC or A→word (and optionally S→ε if empty strings are allowed). Example: VP→Vt NP PP becomes VP→Vt X; X→NP PP. The new intermediate symbol X is introduced; probabilities are distributed so the original derivation probability is preserved.

Span	Candidate constituent	Viterbi score (illustrative)
[0,1]	Det(the)	1.0
[1,2]	N(student)	0.8
[0,2]	NP→Det N	0.60×1.0×0.8=0.48
[2,3]	V(submits)	0.9
[3,4]	Det(the)	1.0
[4,5]	N(form)	0.7
[3,5]	NP→Det N	0.60×1.0×0.7=0.42
[2,5]	VP→V NP	0.50×0.9×0.42=0.189
[0,5]	S→NP VP	0.80×0.48×0.189=0.072576

This is a compact CKY score trace for “the student submits the form”; the chart cell indices are token boundaries. A full implementation stores all candidate non-terminals and split points.

9.4 PCFG limitations

- Context-freeness: rule probability depends on the parent category, not the wider discourse; subject–verb agreement may be missed unless features are added.
- Position invariance: the same VP rule receives the same probability in every context, even when the subject or topic changes.
- Tree-size bias: multiplying many probabilities below 1 can favour shorter trees, even when a longer analysis is linguistically correct.
- Limited lexical sensitivity: a PCFG may not distinguish “with a key” as an instrument from “with a student” as an association unless lexicalised probabilities/features are used.

TASK 10 – IMPLEMENT AND TEST THE PARSERS (3 MARKS)

10.1 Python implementation

The following compact implementation demonstrates a chart-based Earley recogniser for a CFG. For the full assignment, run the top-down parser, feature checks and PCFG CKY scorer on the same 30-sentence set; record measured counts rather than estimating them.

```
from collections import defaultdict
def earley(tokens, grammar, start='S'):
    n=len(tokens); chart=[set() for _ in range(n+1)]
    chart[0].add((start+'^', (start,), 0))
    # grammar: lhs -> list of rhs tuples; '^' is augmented start
    for i in range(n+1):
        agenda=list(chart[i])
        while agenda:
            lhs,rhs,dot,origin = unpack(agenda.pop())
            if dot < len(rhs) and rhs[dot] in grammar:
                B=rhs[dot]  # PREDICTOR
                for prod in grammar[B]: add(chart[i], (B,prod,0,i), agenda)
            elif dot < len(rhs):  # SCANNER
                if i<n and rhs[dot]==tokens[i]:
                    chart[i+1].add((lhs,rhs,dot+1,origin))
            else:  # COMPLETER
                for st in list(chart[origin]):
                    if next_symbol(st)==lhs: advance_into(chart[i],st)
    return any(is_complete_start(x,n) for x in chart[n]), chart
```

Implementation note: the pseudocode above uses helper routines (unpack, add, next_symbol, advance_into, is_complete_start) to make state fields explicit. A runnable implementation must define these helpers and deduplicate states by (lhs, rhs, dot, origin). NLTK's ChartParser/EarleyChartParser can also be used, but document version and grammar.

10.2 Test set (30 sentences)

ID	Test sentence	Phenomenon
1	The student submits the form.	Declarative
2	The students submit the forms.	Declarative
3	I check the timetable.	Declarative
4	They book a room.	Declarative
5	The portal crashed.	Declarative
6	The teacher explains that the exam is postponed.	Complement clause
7	Please print the receipt.	Imperative
8	Submit the scholarship form.	Imperative
9	Book a room near the library.	PP attachment
10	Show the students with the late fee notice.	Ambiguous PP
11	Does the student pay the fee?	Yes/no question
12	Can they book a hostel room?	Yes/no question
13	Where is the library?	Wh-question
14	When will the exams begin?	Wh-question
15	Which room did the student book?	Wh-object
16	Who submitted the form?	Wh-subject
17	The exam that was postponed starts Monday.	Relative clause
18	Tell me whether the hostel is open.	Embedded clause
19	Send me the receipt.	Ditransitive
20	Apply for a hostel room.	PP complement
21	The students submits the form.	Agreement error
22	These student books a room.	Determiner-number error
23	Me submit the form.	Pronoun case error
24	The portal crashed the form.	Subcategorization error
25	*Submit.	Missing object in transitive reading
26	The timetable and exam notice are online.	Coordination
27	The student in the library reads the notice.	PP modifier
28	The teacher will explain the result tomorrow.	Modal + adjunct
29	The student who booked the room submitted the form.	Relative clause
30	The student submits the form near the library after class today.	Long sentence / multiple PPs

10.3 Evaluation results template

Parser	Parse found (%)	Avg parses / sentence	States or calls	Runtime (ms)
Top-down	Measure	Measure	Measure	Measure
Earley	Measure	Measure	Measure	Measure
PCFG / CKY	Measure	Best parse + score	Measure	Measure

For each parser, report success/failure, number of parses, chart states or recursive calls and elapsed time. Include left recursion, PP ambiguity, malformed inputs and sentence lengths up to 20 words. Confirm that the hand-worked Task 7 chart and Task 9 best parse agree with program output.

Plot sentence length on the x-axis and generated states/calls on the y-axis for top-down and Earley. Use the same grammar and test inputs; report the machine, Python version, parser implementation and whether grammar compilation time is included.

TASK 11 – COMPARE, RECOMMEND AND DEFEND THE PARSING DESIGN (2 MARKS)

Criterion	Top-down recursive descent	Earley	PCFG / CKY
Coverage	Depends on grammar; left recursion can loop	General CFG recognition	CNF CFG; selects highest-probability tree
Ambiguity	Backtracking may enumerate many trees	Stores alternatives in chart/packed forest	Ranks trees by probability
Cost	Can be exponential with naïve backtracking	$O(n^3)$ worst-case time, $O(n^2)$ chart space	$O(n^3)$ time for binary CNF CKY
Maintenance	Simple grammar but recursion/order sensitive	Works with left recursion	Requires CNF conversion and probability estimates
Output	One or more derivations	All parses / forest	Best parse and score

Recommendation: deploy a feature-based CFG with Earley parsing for robust recognition and a PCFG/Viterbi ranking layer to select the most plausible parse. Feature unification checks agreement and subcategorization; Earley avoids left-recursion loops and shares partial work; PCFG resolves PP attachment ambiguity. Use a fallback clarification response when several parses have similar scores.

Manual effort includes writing at least 40 rules, curating a 60+ word lexicon, annotating 20+ trees, testing 30 sentences and reviewing errors. A hand-written grammar is economical for a narrow, stable domain. As vocabulary, constructions and input volume grow, maintaining rules and annotation can cost more than using a statistical parser trained on a larger treebank. Hybrid deployment retains domain constraints while reducing manual rule burden.

TASK 12 – DAY-4 DESIGN CHALLENGE: RE-EVALUATE AND ADAPT (8 MARKS)

12A. Impact analysis (2 marks)

The new helpdesk receives noisy conversational multi-clause messages, embedded questions, relative clauses, coordination, fragments, missing punctuation and at least 20% unseen constructions. Original assumptions of controlled vocabulary, fully grammatical input, low ambiguity and manually complete rule coverage are no longer valid.

New construction	Original grammar gap	Likely impact
Embedded wh-clause	Tell me when the exams will be held	No rule for embedded WHQ/CP with subordinate word order.
Relative clause	the exams that were postponed	Relative clause may lack agreement/gap features.
Coordinated clauses	…and whether the hostel fee is due	Clause coordination and mixed clause types need rules.
Ellipsis/fragments	"Hostel fee?" / "And placements?"	No complete S or recoverable missing arguments.
No punctuation/noisy speech	Run-on or disfluent input	Sentence segmentation and parse ambiguity increase.

Longer sentences increase chart positions approximately linearly with token count, while the number of possible spans grows quadratically and general CFG parsing can require cubic time. Ambiguity may grow faster when each PP or clause creates multiple attachment choices.

12B. Evaluation of original design (2 marks)

New message	Original grammar/parser likely behaviour	First failure
Hi, can you tell me when the exams that were postponed will be held?	Greeting and embedded wh-clause are not licensed; parser may fail before reaching main request.	Sentence-level rules / CP.
The hostel fee is due, and can you tell me whether the library is open?	Mixed declarative + coordinated embedded question is outside simple coordination rules.	Clause coordination.
Exams postponed? Need timetable and hostel fee also.	Fragments, omitted auxiliaries and missing punctuation prevent a complete S parse.	Input normalisation / fragment grammar.

Run each message through the original grammar and record the first chart position where no continuation is possible. A top-down parser may backtrack repeatedly; Earley will terminate and return no complete parse if the grammar lacks the construction. A robust partial parser or dependency parser may recover more content economically.

12C. Adaptation or redesign (2 marks)

Decision: MODIFY the hybrid rather than retain the narrow grammar unchanged or replace all components. Extend grammar coverage and add a robust fallback.

Layer	Modification	Expected effect
Preprocessing	Sentence segmentation, punctuation restoration, disfluency handling and greeting detection.	Improves input boundaries.
Grammar	Add embedded WH/YN CP, clause coordination, relative clauses, fragments and optional subject/auxiliary patterns.	Raises coverage.
Features	Thread WH-gap, agreement, tense, case and SUBCAT features through embedded clauses.	Constrains long-distance dependencies.
Parser	Earley for CFG core; partial/fallback parsing for uncovered fragments; packed forest for ambiguity.	Avoids loops and preserves partial meaning.
Probability	Larger domain treebank; lexicalised/parent annotation and smoothing for PCFG.	Improves ranking and attachment.
Feedback	Ask clarification for low-confidence or conflicting parses.	Avoids incorrect helpdesk action.

The revised grammar should be tested on a new conversational treebank, not only the original 30 controlled sentences. Measure exact parse coverage, partial parse coverage, ambiguity, chart size, latency and intent/slot accuracy. Rule additions increase grammar maintenance; a statistical fallback reduces the need to hand-code every conversational variant.

12D. Evidence-based defence (2 marks)

Modification follows from the failure analysis: the original feature CFG + Earley + PCFG remains useful for well-formed controlled queries, but the new 20% unseen-construction rate makes a grammar-only solution brittle. Embedded clauses and coordination should be explicitly added because they are common and structurally meaningful; fragments and noisy text should use a robust partial parser.

Compare original and modified systems on a held-out conversational test set. Report complete parse coverage, partial coverage, exact-match parse accuracy, destination/service slot accuracy, average chart states and p95 latency. A reasonable deployment gate is that the modified system must improve coverage without reducing slot accuracy; choose numerical thresholds from service requirements rather than inventing them.

Remaining limitations include ambiguity in attachment, incomplete semantic/pragmatic interpretation, noisy ASR transcripts and feature-grammar complexity. Keep confidence-aware clarification and human escalation for high-impact fee/examination enquiries.

FINAL DESIGN RECOMMENDATION

Deploy a layered hybrid: normalise conversational input; parse the controlled core with a feature-based CFG and Earley chart parser; rank complete trees with a smoothed PCFG; use partial parsing for fragments/uncovered constructions; and request clarification when the top parses disagree on key helpdesk slots. This balances grammatical control, ambiguity handling and robustness.

APPENDIX A – FINAL SUBMISSION CHECKLIST

Item	Submission check
Grammar	☐ 40+ CFG rules and 60+ lexical entries.
Query set	☐ 30 sentences across sentence types; 5+ ambiguous.
Derivations	☐ Five leftmost/rightmost derivations and tree measures.
Agreement	☐ Six grammatical and six ungrammatical tests.
Subcategorization	☐ Five frames and over/under-complement tests.
Parsing	☐ Top-down traces and full Earley chart.
Features	☐ AVMs, three unifications and feature grammar.
PCFG	☐ 20+ treebank trees, probabilities and CKY table.
Implementation	☐ Working code, parser metrics and plot.
Day-4	☐ Three new messages and measured modified-system results.
Final	☐ Replace all illustrative values/placeholders with actual outputs.

APPENDIX B – FORMULA SUMMARY

Concept	Formula
CFG	$G=(N,\Sigma,R,S)$
PCFG MLE	$P(A\rightarrow\beta\|A)=\text{Count}(A\rightarrow\beta)/\Sigma_\gamma\text{Count}(A\rightarrow\gamma)$
Tree probability	$P(T)=\Pi_r P(\text{rule}_r)$
Sentence probability	$P(w)=\Sigma_{T\in\text{Parses}(w)} P(T)$
Unification	Compatible AVMs merge; conflicting values fail.
Earley state	$[A\rightarrow\alpha\bullet\beta, \text{origin}]$
Earley complexity	$O(n^3)$ time; $O(n^2)$ chart space
CKY Viterbi	$\text{best}[i,j,A]=\max_{k, B,C} P(A\rightarrow BC)\text{best}[i,k,B]\text{best}[k,j,C]$

End of Assignment 3 – CO3.